# Why a Perfect AUC Is Insufficient: Bandwidth-Dependent Margin Loss in Ultra-Wideband Microwave Bone-Fracture Detection

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***Abstract*—Simulation studies of microwave fracture detection routinely report near-perfect classification. We show that such a figure can be uninformative, and that reporting it alone can support a materially wrong engineering decision. Using a full-wave finite-difference time-domain model of a layered forearm, we generated 192 ultra-wideband (3–11 GHz) acquisitions spanning eight simulated specimens, four soft-tissue overlay thicknesses and six fracture states, and scored each under three independent instrument-error realisations. Fractured and intact limbs separate by 40.8 dB with no overlap, giving an area under the receiver operating characteristic curve (AUC) of 1.0000. Because that statistic is saturated, we re-scored the identical 192 cells over restricted frequency windows, a controlled manipulation requiring no further electromagnetic simulation. Removing 3.00–4.35 GHz reduced the class separation from 40.8 dB to 17.9 dB, more than half, *while the AUC remained exactly 1.0000*. The metric registered none of the loss. We identify the mechanism as a loss of range resolution, 3.67 mm to 4.42 mm, which prevents migration from collapsing a layered-interface artefact and converts it into a specimen-dependent false alarm. We argue that any study reporting a saturated classification statistic should report the underlying margin in decibels alongside it, and we quantify what that practice would have changed here.**

***Index Terms*—microwave imaging, ultra-wideband, bone fracture, FDTD, detection margin, area under curve, model evaluation**

## I. Introduction

Fracture diagnosis is dominated by projection radiography and computed tomography, both of which deliver ionising radiation and require fixed clinical infrastructure. This makes them poorly suited to repeated monitoring of healing, and unavailable at the point of injury. Microwave imaging is an attractive alternative because bone and the surrounding soft tissue differ substantially in permittivity, and because a fracture filled with blood presents a strong dielectric discontinuity against cortical bone [1].

The physical difficulty is that the water content creating the contrast also creates the loss. The interrogating signal must cross skin, fat and muscle twice, and attenuation rises steeply with frequency. The usable band is therefore a compromise between range resolution, which improves with bandwidth, and penetration, which degrades with it.

Simulation is the natural first instrument for exploring that compromise, and a large body of such work reports excellent discrimination between fractured and intact bone. This paper is concerned with how those results are *evaluated* rather than with whether they are achievable. Our starting observation is that a classifier which separates two classes completely produces an AUC of exactly 1.0000, and that this value is a *ceiling*. Once reached, the statistic cannot distinguish a system with an enormous margin from one that is close to failing. Any subsequent degradation, whether from reduced bandwidth, a harsher error model or a more difficult anatomy, is invisible until it becomes severe enough to cause an actual misclassification.

We make three contributions.

1) A full-wave study of ultra-wideband fracture detection comprising 192 acquisitions across a factorial design, analysed under a statistical plan fixed before the data were examined, including leave-one-specimen-out validation and a bootstrap that resamples specimens rather than measurements.
2) A controlled demonstration that a saturated AUC conceals the loss of more than half the detection margin. Because the manipulation is applied to the stored full-band response of the same solved cells, bandwidth is varied with everything else held exactly constant.
3) Identification of the mechanism, and a concrete reporting recommendation whose effect on our own engineering decisions we quantify.

## II. Method

### A. Tissue model

Permittivity and conductivity follow the Gabriel four-Cole–Cole parameterisation [1]. A time-domain solver cannot integrate a Cole–Cole response directly, because the fractional-power relaxation has no finite-order time-domain equivalent,

TABLE I
SEPARATION OF FORWARD AND INVERSE MODELS

| | **Forward (data)** | **Inverse (image)** |
|---|---|---|
| Physics | dispersive FDTD | homogeneous migration |
| Tissue | 6 layers, Debye | one scalar $\epsilon_{\mathrm{avg}}$ |
| Geometry | known kerf, cortex | not supplied |
| Grid | 0.2 mm FDTD cells | independent grid |
| Error | injected | blind |

so each tissue is refitted to a two-pole Debye model over 3–11 GHz. The worst deviation across all seven tissues is 0.23 %, and every relaxation time is checked to exceed the solver time step.

### *B. Phantom and experiment design*

The phantom is a layered forearm: skin (2.0 mm), fat (3.0 mm), muscle (1.0 mm), near cortical wall (4.0 mm), marrow (10.0 mm) and far cortical wall (4.0 mm), with the antenna at a 12 mm standoff. A soft-tissue overlay of 0, 3, 6 or 10 mm spans anatomical variation. The fracture is a rectangular blood-filled kerf in the near cortex of width 0.2, 0.4, 1.0 or 2.0 mm, with an additional oblique state.

Eight specimens are drawn from a variation model perturbing cortical wall thickness and overlay, giving a factorial matrix of $8 \times 4 \times 6 = 192$ acquisitions. Without this between-specimen variation the leave-one-specimen-out analysis would be vacuous, because there would be nothing to leave out.

### *C. Forward model and the inverse crime*

If the model generating the data and the model reconstructing the image share assumptions, reconstruction always succeeds and the result is circular. Table I lists the structural separations used to avoid this. The reconstructor receives a matrix of complex $S_{11}$ values and nothing else: not the permittivity profile, the bone geometry, the fracture position or the noise realisation.

Data are generated with gprMax [2] on a 0.2 mm grid in two dimensions ($\mathrm{TM}_z$). An independent analytic layered-medium model, fast enough for development, is used only to build and unit-test the processing chain; every result reported here comes from the full-wave solver.

### *D. Processing chain*

Figure 1 shows acquisition through detection. The antenna steps along the bone axis through 61 positions at 1 mm spacing, transmitting and receiving at each, giving one reflection coefficient per position per frequency. Empty-scene subtraction removes the antenna response; a selective singular-value decomposition removes the layered clutter; Kirchhoff migration focuses the residual scattering back to its origin; and a scalar peak-over-noise metric scores the result in decibels.

One parameter required revision. A common recommendation is to discard the first two or three singular components. Measured on two independent targets, this destroys the fracture signature along with the clutter. We instead discard a component only if, after migration, its energy spans more than half the scan aperture, which is the signature of a layered interface rather than a localised scatterer.

TABLE II
DETECTOR SCORE DISTRIBUTIONS, 576 SCORES

| **Class** | **n** | **Min** | **Median** | **Max** |
|---|---|---|---|---|
| Intact | 96 | 8.8 | 19.3 | **27.7** |
| Fractured | 480 | **68.5** | 89.9 | 104.8 |

### *E. Instrument error*

Receiver noise, cable phase drift, stage jitter and standoff variation are injected *after* the forward solve. One electromagnetic solution therefore yields several independent "remounts", which is both cheaper and physically correct, since a remount perturbs the instrument rather than the anatomy. Three realisations per acquisition give 576 scores.

### *F. Statistical analysis plan*

The plan was fixed before the data were examined. Two points require care. The detector has no fitted parameters, so the only trained quantity is the decision threshold; leave-one-specimen-out validation is therefore applied to the threshold rather than to the detector. The bootstrap resamples *specimens*, not measurements, because twenty-four acquisitions from one bone are not twenty-four independent observations and resampling them would give an interval several times too narrow. Permutation labels are shuffled within specimen for the same reason.

## III. RESULTS

### *A. Detection performance*

Table II gives the raw distributions. The classes are separated by **40.8 dB with no overlap**, a factor of 110 in amplitude. The AUC is 1.0000 at every overlay thickness and every fracture width, with a bootstrap 95 % interval over specimens of 1.000–1.000 and a permutation $p$ of 0.001, the floor for 1000 shuffles. Leave-one-specimen-out validation of the threshold gives sensitivity 0.9979 (Wilson 0.988–1.000) and specificity 1.0000 (0.962–1.000). Repeatability across remounts is ICC(1,1) $= 0.9979$.

A perfect AUC ordinarily indicates leakage between training and test data rather than a genuine result, and was treated as such. Inspection of the distributions excludes that explanation: the gap is real, and its structure follows physical prediction. The weakest fractured cases are the narrowest kerf under the thickest overlay, and the strongest intact cases also occur at the thickest overlay. A leak would not order itself by physical difficulty.

### *B. Bandwidth-dependent margin loss*

Every solved cell stores the full-band response, so the entire matrix can be re-scored over a restricted window with no further electromagnetic simulation. Bandwidth is thus varied with the geometry, the error model, the processing chain and the detector all held exactly constant.

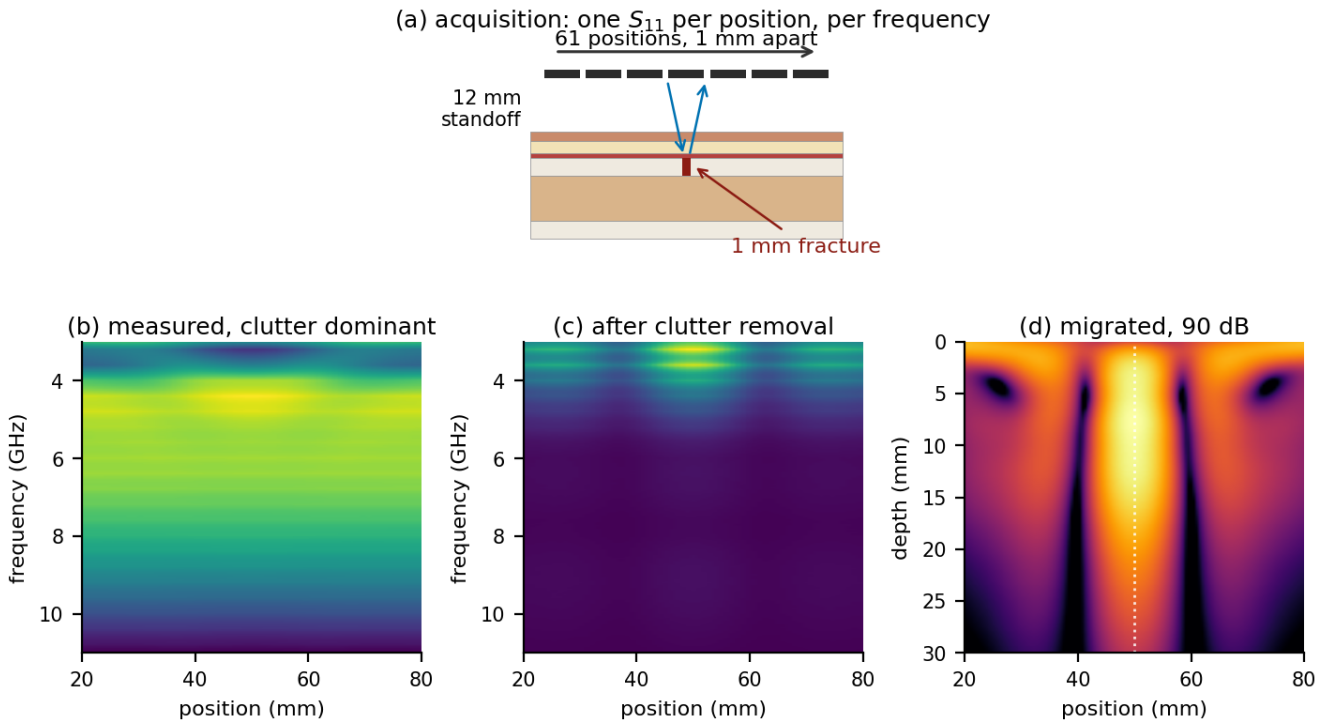



Fig. 1. Acquisition and processing. (a) The antenna steps along the bone axis at 12 mm standoff, transmitting and receiving at each of 61 positions. (b) The measured response after empty-scene subtraction, dominated by layered clutter; the fracture is present and invisible. (c) After selective singular-value removal, the local scattering remains. (d) Migration focuses it to the true position (dotted). Panels (b)–(d) are solver output for a 1 mm fracture under 6 mm of overlay.

TABLE III
THE SAME 192 CELLS, RE-SCORED OVER RESTRICTED BANDS

| Window | BW | Res. | AUC | Sep. |
|---|---|---|---|---|
| 3.00–11.0 GHz | 8.00 GHz | 3.67 mm | 1.0000 | **40.8 dB** |
| 4.35–11.0 GHz | 6.65 GHz | 4.42 mm | 1.0000 | **17.9 dB** |
| 5.00–11.0 GHz | 6.00 GHz | 4.90 mm | 1.0000 | 16.0 dB |

Table III and Fig. 2 give the result. Removing 3.00–4.35 GHz costs **22.9 dB, more than half the total separation**. The AUC is 1.0000 in all three windows and registers none of it.

### C. Mechanism

Band limitation degrades both classes, and the intact class is the more damaging. The weakest fractured score falls from 68.5 to 55.7 dB, as expected from reduced penetration and resolution. The strongest *intact* score *rises* from 27.7 to 37.8 dB, which is a false alarm rather than a missed detection. It occurs on one specimen at 6 mm overlay against a group mean of 23.7 dB.

The mechanism follows from the resolution column of Table III. Bandwidth buys range resolution. At full bandwidth, migration resolves that specimen's layered-interface reflection and collapses it; band-limited, the artefact smears into the detection window and registers as a target. Because the effect is specimen-dependent, the concern is variance rather than the mean shift: one specimen in eight spikes, and a larger cohort would be expected to contain worse outliers.

### D. Consequence for the instrument

This measurement was not academic. An antipodal Vivaldi antenna [4] designed in openEMS [3] initially matched 4.35–12.00 GHz against a 3–11 GHz target, and the shortfall had been documented as an acceptable compromise on the strength of an unchanged AUC. Table III shows that compromise cost 22.9 dB. Re-examining the input impedance showed the shortfall was never a low-frequency cutoff: the design was already matched at 2.225–2.725 GHz, reaching −33 dB, two gigahertz below its stated band edge. What lay between was a mismatch, a resistive plateau of 24–37 Ω across 3.25–4.40 GHz. A load of 24.4 Ω into a 50 Ω feed predicts −9.3 dB return loss and −9.14 dB was measured.

A tapered feed, 4.5 mm at the flare and 1.74 mm at the port, moved the match to 2.48–12.00 GHz, covering 99.69 % of the design band with a worst in-band value of −9.98 dB and a median input resistance of 45.6 Ω. Peak gain is 7.02, 11.87, 12.49, 12.06 and 9.33 dBi at 3, 5, 7, 9 and 11 GHz respectively; the 8–10 dBi target is met from approximately 4 GHz upward and is not met at 3 GHz, where the aperture is 0.6 wavelengths.

### E. Exposure

The band straddles a boundary in current guidelines: specific absorption rate applies below 6 GHz and absorbed power density above it [5], [6]. Both were evaluated for a layered

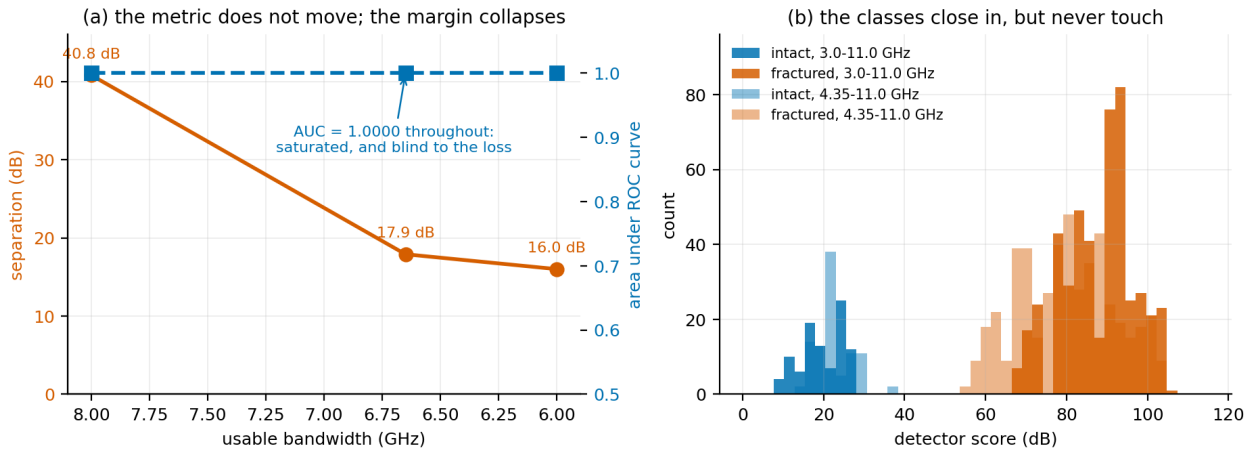



Fig. 2. (a) As usable bandwidth falls, the class separation collapses from 40.8 dB to 16.0 dB while the AUC remains pinned at 1.0000. (b) The underlying distributions converge from both sides but do not overlap, which is precisely why the classification statistic cannot see the change.

limb at 12 mm standoff, with tissue densities from [7]. At the specified 0 dBm transmit power and worst-case illumination, all limits pass with a minimum margin of 18×; the binding quantity is absorbed power density above 6 GHz.

## IV. Discussion

A saturated statistic is a lower bound, not a measurement. The defensible claim from Table II is that under the modelled error budget the classes separate by 40.8 dB with no overlap, not that detection is perfect. The binding assumption is the error model rather than the physics: a physical bench adds cable flex, connector repeatability, operator variation and genuine inter-specimen heterogeneity, none of which are simulated. The 40.8 dB margin is the budget those unmodelled terms must consume before the conclusion changes, and that framing is only available because the margin was reported.

The practical consequence is concrete. Judged by AUC alone, the initial antenna was adequate: the statistic did not move. Judged by margin, it cost more than half the available budget, which is what motivated the redesign reported above. The same reasoning applies to any design decision evaluated against a saturated metric, including grid refinement, averaging time and dynamic range.

We therefore suggest that studies reporting near-perfect classification also report the separation between classes in the detector's own units, together with the extreme values of each class. This costs nothing, is not an additional experiment, and distinguishes a comfortable result from a marginal one in exactly the regime where the classification statistic cannot.

## V. Limitations

1) The injected error model is the binding assumption. A physical bench adds terms this work does not model.
2) Every fracture is at a known position; the study measures separability, not localisation.
3) Propagation is two-dimensional $TM_z$. A three-dimensional cylindrical model has been implemented and verified against the solver but the comparison is not yet run, so no claim is made about the effect of curvature or out-of-plane scattering.
4) Eight simulated specimens are not eight cadaveric bones.
5) The imaging model excites with an elementary dipole, not the antenna of Section III-D; antenna and imaging results are independent.
6) Antenna gain at 3 GHz is unmet, and was shown not to be reachable by resizing within the specified board.
7) Exposure is assessed by calculation, which is not a substitute for measurement to IEC 62209.

No hardware was built and no human or animal subject was exposed at any point in this work.

## VI. Conclusion

Across 192 full-wave acquisitions from eight simulated specimens, fractured and intact limbs separate by 40.8 dB with no overlap, giving an AUC of 1.0000 with leave-one-specimen-out sensitivity 0.998 and specificity 1.000. Restricting the usable band removed more than half of that separation while the AUC remained exactly 1.0000, and the mechanism is a loss of range resolution that converts a resolvable layered-interface artefact into a specimen-dependent false alarm. A perfect classification score is therefore a ceiling rather than a result, and should be accompanied by the margin it conceals.

All code, data-generating scripts and figures are available at https://github.com/amalnathkp13-boop/bone-fracture-rf-sensing.

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